THE SIMPLICIAL RESOLUTION OF THE CONSTANT SHEAF Z ON THE TOTAL SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the argument is parallel to the ringed case delivered as card L46181 (lemma-simplicial-resolution-ringed), with O replaced by the constant sheaf Z; the g_{n!} description is from card L46149 (lemma-restriction-to-components-site); no verification is omitted.

Editorial note. The label lemma-simplicial-resolution-Z-site names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The simplicial-site situation and the description of g_{n!}. Same chapter, Situation \label{situation-simplicial-site} and \label{lemma-restriction-to-components-site} (card L46149). Statement lines 1711-1714 and proof lines 1721-1723: the functors g_{n!} and their description from the restriction lemma, used without further mention.
- The free abelian sheaf on a simplicial set and homotopy invariance. Simplicial, Example \label{simplicial-example-simplex-simplicial-set}, Example \label{simplicial-example-simplex-contractible}, Remark \label{simplicial-remark-homotopy-better} and \label{simplicial-lemma-homotopy-equivalence-s-N}. Proof lines 1730-1749: g_m^{-1} of the complex is the free abelian sheaf on Delta[m]; Delta[m] is homotopy equivalent to Delta[0] and taking free abelian sheaves is functorial.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 1723; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 08Q3: sources/stacks-chow/tags.txt, source lines 11046–11046. Prerequisite pointer invoked at proof line(s) 1741; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 0176: sources/stacks-chow/tags.txt, source lines 1569–1569. Prerequisite pointer invoked at proof line(s) 1739; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 019T: sources/stacks-chow/tags.txt, source lines 1664–1664. Prerequisite pointer invoked at proof line(s) 1746; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 019M: sources/stacks-chow/tags.txt, source lines 1657–1657. Prerequisite pointer invoked at proof line(s) 1745; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 1712; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE and with notation as above there is a complex*

$$\ldots \to g_{2!}\mathbf{Z} \to g_{1!}\mathbf{Z} \to g_{0!}\mathbf{Z}$$

of abelian sheaves on $\mathcal{C}_{total}$ which forms a resolution of the constant sheaf with value $\mathbf{Z}$ on $\mathcal{C}_{total}$.

Proof. We will use the description of the functors $g_{n!}$ in Lemma 09WG without further mention. As maps of the complex we take $\sum(-1)^i d_i^n$ where $d_i^n : g_{n!}\mathbf{Z} \to g_{n-1!}\mathbf{Z}$ is the adjoint to the map $\mathbf{Z} \to \bigoplus_{[n-1]\to[n]} \mathbf{Z} = g_n^{-1}g_{n-1!}\mathbf{Z}$ corresponding to the factor labeled with $\delta_i^n : [n-1] \to [n]$. Then g_m^{-1} applied to the complex gives the complex

$$\ldots \to \bigoplus\nolimits_{\alpha\in \mathrm{Mor}_\Delta([2],[m])} \mathbf{Z} \to \bigoplus\nolimits_{\alpha\in \mathrm{Mor}_\Delta([1],[m])} \mathbf{Z} \to \bigoplus\nolimits_{\alpha\in \mathrm{Mor}_\Delta([0],[m])} \mathbf{Z}$$

on $\mathcal{C}_m$. In other words, this is the complex associated to the free abelian sheaf on the simplicial set $\Delta[m]$, see Simplicial, Example 0176. Since $\Delta[m]$ is homotopy equivalent to $\Delta[0]$, see Simplicial, Example 08Q3, and since "taking free abelian sheaf on" is a functor, we see that the complex above is homotopy equivalent to the free abelian sheaf on $\Delta[0]$ (Simplicial, Remark 019M and Lemma 019T). This complex is acyclic in positive degrees and equal to $\mathbf{Z}$ in degree 0. □